

Use of Endocrine Therapy for Breast Cancer Risk Reduction: ASCO Clinical Practice Guideline Update Summary

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ASCO first published clinical practice recommendations for the use of pharmacologic interventions for breast cancer risk reduction in 1999.¹ ASCO updated these breast cancer risk reduction guideline recommendations in 2002, 2009, and 2013.²⁻⁴ The current update was undertaken to integrate data from the International Breast Intervention Study II (IBIS-II) randomized, placebo-controlled trial on the use of anastrozole (1 mg per day orally for 5 years) to reduce the incidence of breast cancer in postmenopausal women at increased risk of developing breast cancer.⁵

The full guideline article summarizes the results of a literature search and review of new data on breast cancer risk reduction published since 2013 and provides new recommendations for the use of anastrozole in breast cancer risk reduction. These new recommendations supplement the guideline recommendations issued in 2013.

In addition, this update, which resulted from a discussion of clinical considerations, addresses some questions frequently raised by patients and/or providers regarding the use of tamoxifen, raloxifene, or an aromatase inhibitor for breast cancer risk reduction. The recent placebo-controlled trial of a lower daily dose of tamoxifen (5 mg) for 3 years in women with intraepithelial neoplasia (atypical hyperplasia or in situ cancer) is also reviewed.⁶ Additional information is available at www.asco.org/breast-cancer-guidelines. Patient information is available at www.cancer.net.

WHAT IS PRACTICE CHANGING?

Anastrozole 1 mg/day is now added to three other endocrine agents (tamoxifen 20 mg, raloxifene 60 mg, and exemestane 25 mg) shown to reduce breast cancer incidence in large randomized trials when prescribed once per day for 5 years to postmenopausal women at increased risk. Concern about adverse effects is a major reason for poor uptake of endocrine therapy for breast cancer risk reduction, so the option of low-dose tamoxifen (5 mg/day) for 3 years may be an alternative option in women with intraepithelial neoplasia.

Since the last guideline update, it has become clear that it is not sufficient to cite the eligibility criteria of randomized prevention trials when discerning who might benefit from endocrine prevention. A better approach is to define a threshold above which women are most likely to benefit from endocrine preventive therapy. Those most likely to benefit from endocrine prevention therapy are women with one or more of the following: a prior diagnosis of atypical hyperplasia or lobular carcinoma in situ, a calculated National Cancer Institute Breast Cancer Risk Assessment Tool [BCRAT]) 5-year risk of at least 3%, a 10-year risk (IBIS/Tyrer-Cuzick Risk Calculator) of 5% or higher, a relative risk of at least two times the average for their age group if age is 45 to 69 years or at least four times the average for their age group if age is 40 to 44 years.

Author affiliations and support information (if applicable) appear at the end of this article.

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THE BOTTOM LINE

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Guideline Question

What is the role of anastrozole in reducing the risk of developing breast cancer in women not previously diagnosed with breast cancer?

Target Population

Women without a personal history of breast cancer who are at increased risk of developing the disease.

Target Audience

Medical oncologists, surgical oncologists, gynecologists, primary care physicians, and general practitioners.

Methods

An Expert Panel was convened to develop updated clinical practice guideline recommendations based on a systematic review of the medical literature.

Updated Recommendations

Recommendation 4.1. Anastrozole (1 mg/day orally for 5 years) should be discussed as an alternative to tamoxifen, raloxifene, or exemestane to reduce the risk of invasive breast cancer in postmenopausal women at increased risk of developing breast cancer.

Recommendation 4.2. Women most likely to benefit are those with one of more of the following: a diagnosis of atypical (ductal or lobular) hyperplasia or lobular carcinoma in situ, an estimated 5-year risk (National Cancer Institute Breast Cancer Risk Assessment Tool [BCRAT]) of at least 3%, a 10-year risk (International Breast Intervention Study [IBIS]/Tyrer-Cuzick Risk Calculator) of at least 5%, or a relative risk of at least four times the population risk for their age group if age is 40 to 44 years or two times that for age group if age is 45 to 69 years.

Recommendation 4.3. Clinicians should not prescribe anastrozole, exemestane, or raloxifene for breast cancer risk reduction in premenopausal women.

Recommendation 4.4. Discussions between patients and health care providers should include both the benefits and risks of anastrozole along with the other approved drugs for risk reduction based on menopausal status.

Recommendation 4.5. Prior to initiating an aromatase inhibitor, clinicians should evaluate patients for baseline fracture risk and measure bone mineral density. Multiple studies have reported an increased rate of bone loss in women treated with aromatase inhibitors. Clinicians should use anastrozole with caution in postmenopausal women with moderate bone mineral density loss, and if it is used, they should consider the use of bone protective agents such as bisphosphonates and receptor activator of nuclear factor kappa beta ligand (RANKL) inhibitors. All patients receiving aromatase inhibitors should be encouraged to exercise regularly and take adequate calcium and vitamin D supplements. A history of osteoporosis and/or severe bone loss is a relative contraindication for the use of anastrozole. In IBIS-II, women with severe osteoporosis (T score < -4 or more than two vertebral fractures) were excluded. Other endocrine preventive therapies that do not reduce bone density, such as tamoxifen or raloxifene, are also available for this group of women.

Recommendation 4.6. Clinicians should also inform women of the possibility of joint stiffness, arthralgias, vasomotor symptoms, hypertension, dry eyes, and vaginal dryness while taking anastrozole. (Type of recommendation: evidence based, benefits outweigh harms; Evidence quality: high; Strength of recommendation: strong).

Refer to [Table 1](#) for the full list of recommendations.

Additional Resources

More information, including a Data Supplement, slide sets, and clinical tools and resources, is available at www.asco.org/breast-cancer-guidelines. Patient information is available at www.cancer.net. The Methodology Manual, available at www.asco.org/guideline-methodology, provides additional information about the methods used to develop this guideline.

ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

TABLE 1. Summary of All Recommendations

Agent	Recommendation	Strength of Recommendation and Strength of Evidence
Tamoxifen	1.1. Should be discussed as an option to reduce the risk of invasive BC, specifically ER-positive BC, in premenopausal women who are age ≥ 35 years with a 5-year projected absolute BC risk of $\geq 1.66\%$ or with LCIS. Risk reduction benefit continues for at least 10 years.	Strong, evidence-based recommendation. Strength of evidence: Strong evidence, based on five RCTs with low risk of bias
	1.2. Is not recommended for use in women with a history of deep vein thrombosis, pulmonary embolus, stroke, or transient ischemic attack or during prolonged immobilization.	
	1.3. Is not recommended for women who are pregnant, women who may become pregnant, or nursing mothers.	
	1.4. Is not recommended in combination with hormone therapy.	
	1.5. Follow-up should include a timely work-up of abnormal vaginal bleeding.	
	1.6. Discussions with patients and health care providers should include both the risks and benefits of tamoxifen in the preventive setting.	
	1.7. Dosage: 20 mg/day orally for 5 years.	
Raloxifene	2.1. Should be discussed as an option to reduce the risk of invasive BC, specifically ER-positive BC, in postmenopausal women who are age ≥ 35 years with a 5-year projected absolute BC risk of $\geq 1.66\%$ or with LCIS.	Strong, evidence-based recommendation. Strength of evidence: Strong evidence, based on four RCTs with low risk of bias
	2.2. May be used longer than 5 years in women with osteoporosis in whom BC risk reduction is a secondary benefit.	
	2.3. Should not be used for BC risk reduction in premenopausal women.	
	2.4. Is not recommended for use in women with a history of deep vein thrombosis, pulmonary embolus, stroke, or transient ischemic attack or during prolonged immobilization.	
	2.5. Discussions with patients and health care providers should include both the risks and benefits of raloxifene in the preventive setting.	
	2.6. Dosage: 60 mg/day orally for 5 years.	
Exemestane	3.1. Should be discussed as an alternative to tamoxifen and/or raloxifene to reduce the risk of invasive BC, specifically ER-positive BC, in postmenopausal women age ≥ 35 years with a 5-year projected absolute BC risk of $\geq 1.66\%$ or with LCIS or atypical hyperplasia.	Moderate, evidence-based recommendation. Strength of evidence: Moderate evidence, based on one RCT with low risk of bias
	3.2. Should not be used for BC risk reduction in premenopausal women.	
	3.3. Discussions with patients and health care providers should include both the risks and benefits of exemestane in the preventive setting	
	3.4. Dosage: 25 mg/day orally for 5 years.	
Anastrozole	4.1. Anastrozole (1 mg/day orally for 5 years) should be discussed as an alternative to tamoxifen, raloxifene, or exemestane to reduce the risk of invasive BC in postmenopausal women at increased risk of developing BC.	Evidence based, benefits outweigh harms; Evidence quality: high; Strength of recommendation: Strong, based on one RCT with low risk of bias
	4.2. Women most likely to benefit are those with one of more of the following: a diagnosis of atypical (ductal or lobular) hyperplasia or lobular carcinoma in situ, an estimated 5-year risk (National Cancer Institute Breast Cancer Risk Assessment Tool [BCRAT]) of at least 3%, a 10-year risk (IBIS/Tyrer-Cuzick Risk Calculator) of at least 5%, or a relative risk of at least four times the population risk for age group if age is 40 to 44 years or two times that for their age group if age is 45 to 69 years.	
	4.3. Clinicians should not prescribe anastrozole, exemestane, or raloxifene for BC risk reduction in premenopausal women	
	4.4. Discussions between patients and health care providers should include both the benefits and risks of anastrozole along with the other approved drugs for risk reduction based on menopausal status.	

(continued on following page)

TABLE 1. Summary of All Recommendations (continued)

Agent	Recommendation	Strength of Recommendation and Strength of Evidence
	4.5. Prior to initiating an aromatase inhibitor, clinicians should evaluate patients for baseline fracture risk and measure bone mineral density. Multiple studies have reported an increased rate of bone loss in women treated with aromatase inhibitors. Clinicians should use anastrozole with caution in postmenopausal women with moderate bone mineral density loss, and if it is used, they should consider the use of bone protective agents such as bisphosphonates and RANKL inhibitors. All patients receiving aromatase inhibitors should be encouraged to exercise regularly and take adequate calcium and vitamin D supplements. A history of osteoporosis and/or severe bone loss is a relative contraindication for the use of anastrozole. In IBIS-II, women with severe osteoporosis (T score < -4 or more than two vertebral fractures) were excluded. Other endocrine preventive therapies that do not reduce bone density, such as tamoxifen or raloxifene, are also available for this group of women.	
	4.6. Clinicians should also inform women of the possibility of joint stiffness, arthralgias, vasomotor symptoms, hypertension, dry eyes, and vaginal dryness while taking anastrozole.	

Abbreviations: BC, breast cancer; ER, estrogen receptor; IBIS, International Breast Intervention Study; LCIS, lobular carcinoma in situ; RANKL, receptor activator of nuclear factor kappa beta ligand; RCT, randomized controlled trial.

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